

2023-Ph-H1-Q25

Section: Electricity

Topic: Monitoring and Measuring a.c.

An a.c. supply has a peak voltage of 14.1 V. What is its r.m.s. voltage?

Answer: B

(10.0 V)

Guidance for Students:

Use the conversion:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{14.1}{\sqrt{2}} \approx 10.0 \text{ V}$$

Revision Tips:

- Peak voltage: max value of waveform
 - R.m.s. = equivalent DC voltage
 - $V_{\text{rms}} \approx 0.707V_{\text{peak}}$
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